Greecon
SH.P.K

12

HARDWARE SHEET HW12 · QUALIFIED BY SPECIFICATION

Flow meter

The meter that says how much water is moving. With the pump's running contact it lets Greecon say that the pump is running but no water is flowing.

QUALIFIES	Qualified by specification. Any flow meter for the pipe and the fluid that has a 4 to 20 mA output.
USED IN	Optional in kits H, F and W (H hotel or guesthouse, S solar plant, F farm or greenhouse, W water tank and pump station)
FOR	The project manager who decides at the survey whether the site gets one, the plumber who fits it and the electrician who wires it
COVERS	How it qualifies · what it does at each kind of site · settings · fitting · in the platform · the acceptance test · what goes wrong · what is not yet proven · the buying record



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HW12

Flow meter

The flow meter says how much water is moving in the pipe, in litres a minute. By itself it gives a number and its trend. Together with the pump's running contact (sheet HW09) it gives the alert a pump station most wants: the pump is running but no water is flowing. It is optional in every kit, and it is the dearest of the transmitters, so the survey decides whether the site gets one.

At a glance

WHAT IT DOES	Measures the flow in the pipe. Without it (or without the pump's running contact) the alert for a pump that runs with no water moving cannot be raised.
HOW IT QUALIFIES	**Qualified by specification.** Any flow meter for the pipe and the fluid that has a 4 to 20 mA output.
SPECIFICATION	4 to 20 mA output. Needed for "the pump is running but no water is flowing"
WIRED TO	Input 3 of the 4 to 20 mA module (sheet HW08). Its manual says whether its output powers the loop itself or needs the 24 V supply.
FITTED BY	A plumber fits it in the pipe, with the straight run its manual asks for. The electrician wires its signal.
PLANNING ESTIMATE (EUR)	150 to 500. Not a quote.
USED IN	Optional in kits H, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

Any make qualifies that meets every line below. An electromagnetic meter and a turbine meter are the usual kinds.

It qualifies if:

- ☐ It suits the pipe and the water: the pipe's size and material, and clean or dirty water. On a drinking-water pipe, its wetted parts are approved for drinking water.
- ☐ **It has a 4 to 20 mA output.** A meter with only a pulse output does not qualify: the module reads current.
- ☐ Its manual says which flow 20 mA stands for (the top of its range), or lets that be set, and that flow is above the most the pump delivers.
- ☐ Its manual says whether its output powers the loop itself, or needs the loop's 24 V supply. The electrician needs to know which.
- ☐ The pipe has the straight run the manual asks for, before and after the meter. Look at the survey.

STOP

Do not buy a flow meter that is not rated for the fluid, or one whose only output is pulses. Fitting it means cutting the pipe: that is a plumber's work, agreed with the customer, with the pump stopped by the customer's own person.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	OPTIONAL	0 or 1	With the pump contact module it makes the no-flow alert possible: the pump is running but no water is flowing.
S Solar plant with grid connection	NOT USED	None	A solar plant has no water side.
F Farm or greenhouse with irrigation	OPTIONAL	0 or 1	Only if there is a pump to watch.
W Water tank and pump station	OPTIONAL	0 or 1	The part behind a pump station's most useful alert: pump running, no flow. Without it the handover says that the alert cannot be raised.

What is lost without it. Without a flow meter, or without the pump's running contact, Greecon cannot say that the pump is running but no water is flowing. The site then ticks **Not fitted at this site**, and the handover says that this alert cannot be raised.

What Greecon reads from it

One reading: **Pump flow**, in litres a minute. 4 mA is no flow, and 20 mA is the top of the range.

The top of the range	4 mA	12 mA	20 mA
100 litres a minute (6 cubic metres an hour)	0	50	100
500 litres a minute (30 cubic metres an hour)	0	250	500

Greecon asks for litres a minute. A range in cubic metres an hour is multiplied by 16.67, and one in litres a second by 60.

Standard alert	Opens when	Kind
No flow	The pump is running (Pump running reads 1) and the flow has been at or below 0.2 litres a minute for 30 seconds	Act now

The alert needs the pump's running state to be read as well. With no pump contact module it is never raised. A flow below zero, or above 100,000 litres a minute, is called a reading that cannot be real.

Settings, and where they are made

What	Value	Where it comes from
The top of the range	The flow that 20 mA stands for, in litres a minute	The meter's label or its manual; on a meter with a menu, what is set there
Its input on the module	Input 3	Wired by the electrician
The no-flow limit	0.2 litres a minute as standard. See the check below	In the platform, with the owner

CHECK **Read the flow with the pump stopped and the pipe full.** A meter at rest seldom sends exactly 4 mA. If Greecon then shows 0.4 litres a minute, a limit of 0.2 would never open the alert. Agree a limit a little above the resting number and far below the pump's ordinary flow, and write both numbers down.

Fitting

A plumber fits the meter in the pipe, and ELECTRICIAN wires its signal. PROJECT MANAGER watches, photographs and writes down.

- ☐ The meter sits in the pipe the pump feeds, with the **straight run** its manual asks for before and after it, and the arrow on its body pointing the way the water goes.
- ☐ Its signal is wired to **input 3** of the module, as the meter's manual and the module's manual draw it.
- ☐ **One supply in the loop, never two.** A meter whose output powers the loop is wired without the 24 V supply; one that needs the loop's supply has the 24 V supply in series, as for the other transmitters (sheet HW08).
- ☐ A meter that needs mains power is connected by the electrician.
- ☐ The wires are numbered at both ends. The label and the terminals are photographed.

In the platform

The flow meter is not a device in the platform. It is one reading of the **Water I/O module**, which the box finds in step 4, **Where each device is on the network**.

- **Its range.** In step 5, **Which register holds which reading**, under **Flow meter**, give the **Top of the range** in litres a minute and press **Save the ranges**. Greecon works out the register and the scale: nobody types either.
- **No flow meter at this site:** tick **Not fitted at this site** under it. The page asks first, because the reading and its history are removed.
- **The limit.** In step 10, **Alert limits agreed with the customer**, the limit is called *No flow at or below*. The standard is 0.2 litres a minute.

Acceptance test

At the office, before the kit leaves:

- ☐ The label is photographed, and the top of its range, in litres a minute, is written on the install card.
- ☐ With a loop calibrator on input 3 in place of the meter, the module reads 4000, 12000 and 20000 (sheet HW08).

```
greecon-edge modbus-test --host 192.168.1.50 --unit 3 --type input --address 2
```

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
Flow. A bucket test (fill a measured volume and time it), or the flow meter's own display.	Within 10 percent			☐ Pass ☐ Fail
At rest. The pump stopped by the owner's own person, the pipe full.	The flow is below the no-flow limit. Write the number down			☐ Pass ☐ Fail

Do not start or stop the pump yourself: the owner's own person does, because they know what else it affects.

What goes wrong

What you see	Likely cause	What to do
A flow far below zero (*-125 cannot be a flow*)	No current in the loop: a wire has come off, the loop has no supply, or the meter has no power. The module reads 0	Read input 3 from the box: 0 means no current. The electrician checks the loop
A flow a little below zero while the pump is stopped (*-0.2 cannot be a flow*)	The meter rests a hair under 4 mA. The box reads about 4000, not 0	Check the reading with the pump running. Write the resting number down and report it to Greecon
The no-flow alert does not open when the pump runs dry	The meter's resting number is above the limit, or the pump's running state is not read	Set the limit above the resting number. Check that Pump running reads 1 while the pump runs (sheet HW09)
The no-flow alert opens while the pump is stopped	The pump's contact is inverted	The electrician uses the starter's other contact (sheet HW09)
The flow is a steady share too high or too low	The top of the range was given in the wrong unit (cubic metres an hour for litres a minute), or is not what the meter is set to	Read the meter's own setting, and give the range again
A flow below zero at a site that has no flow meter	The reading was left in place, and an empty input reads like a wire off	Tick **Not fitted at this site** under the flow meter

More is in the runbooks (Project Manager Field Manual, PDF 02): R3 (a reading looks wrong) and R18 (a reading is impossible after the ranges were given).

Not yet proven

- This software has not yet read a flow meter through a real 4 to 20 mA module. The scale and the no-flow alert are tested against pretend equipment.
- **How a real meter rests is not known.** The module reads in small steps, and a meter at rest may sit a little above or below 4 mA. That is why the resting number is read and written down at commissioning, and why the limit is agreed after it.
- The prices are planning figures, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	